Wave Reflection and Transmission at Oblique Incidence: Perpendicular Polarization – Part 1

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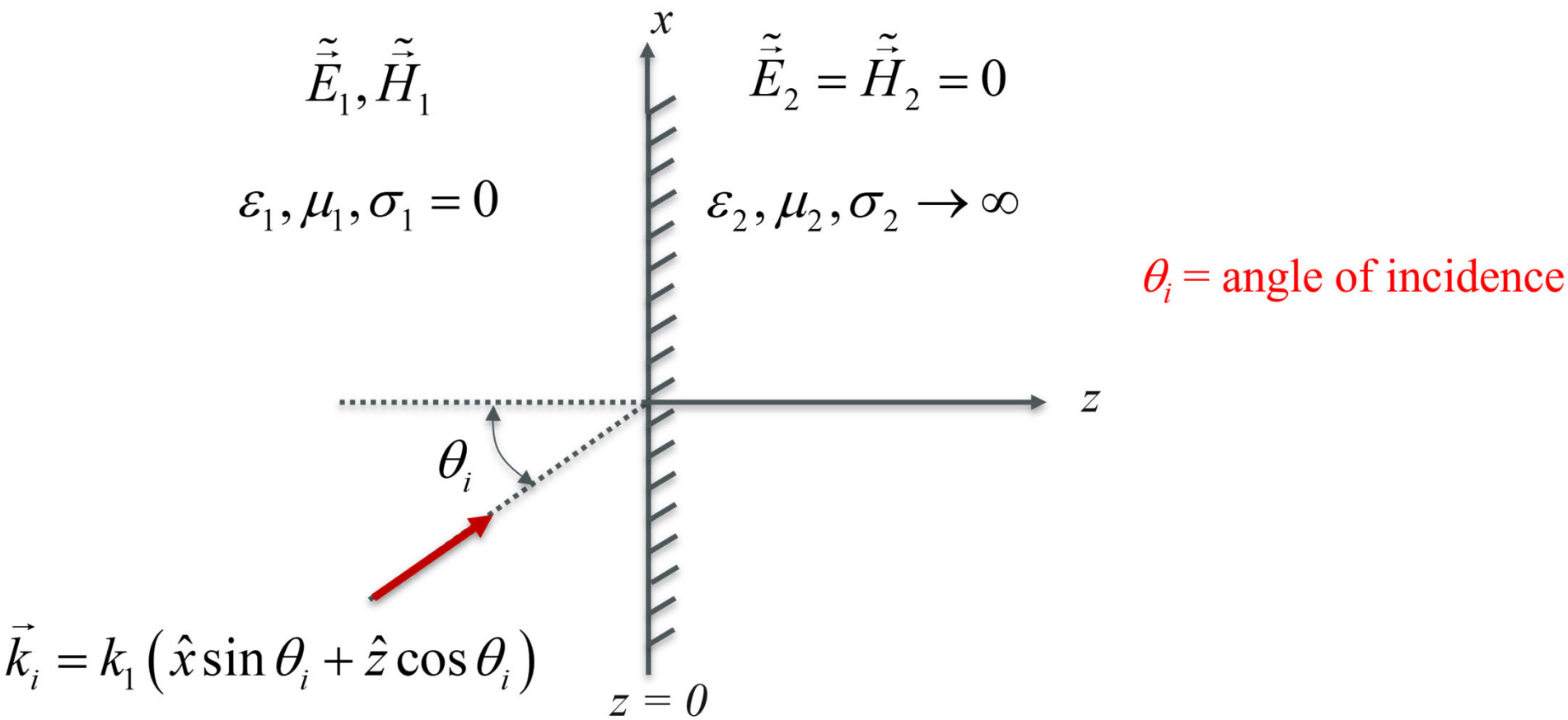
Lecture Objective

By the end of this lecture, you will be prepared to:

- Analyze plane wave reflection from a perfectly conducting boundary at oblique incidence for a wave with perpendicular polarization.

Oblique Incidence On a Plane Perfectly Conducting Boundary



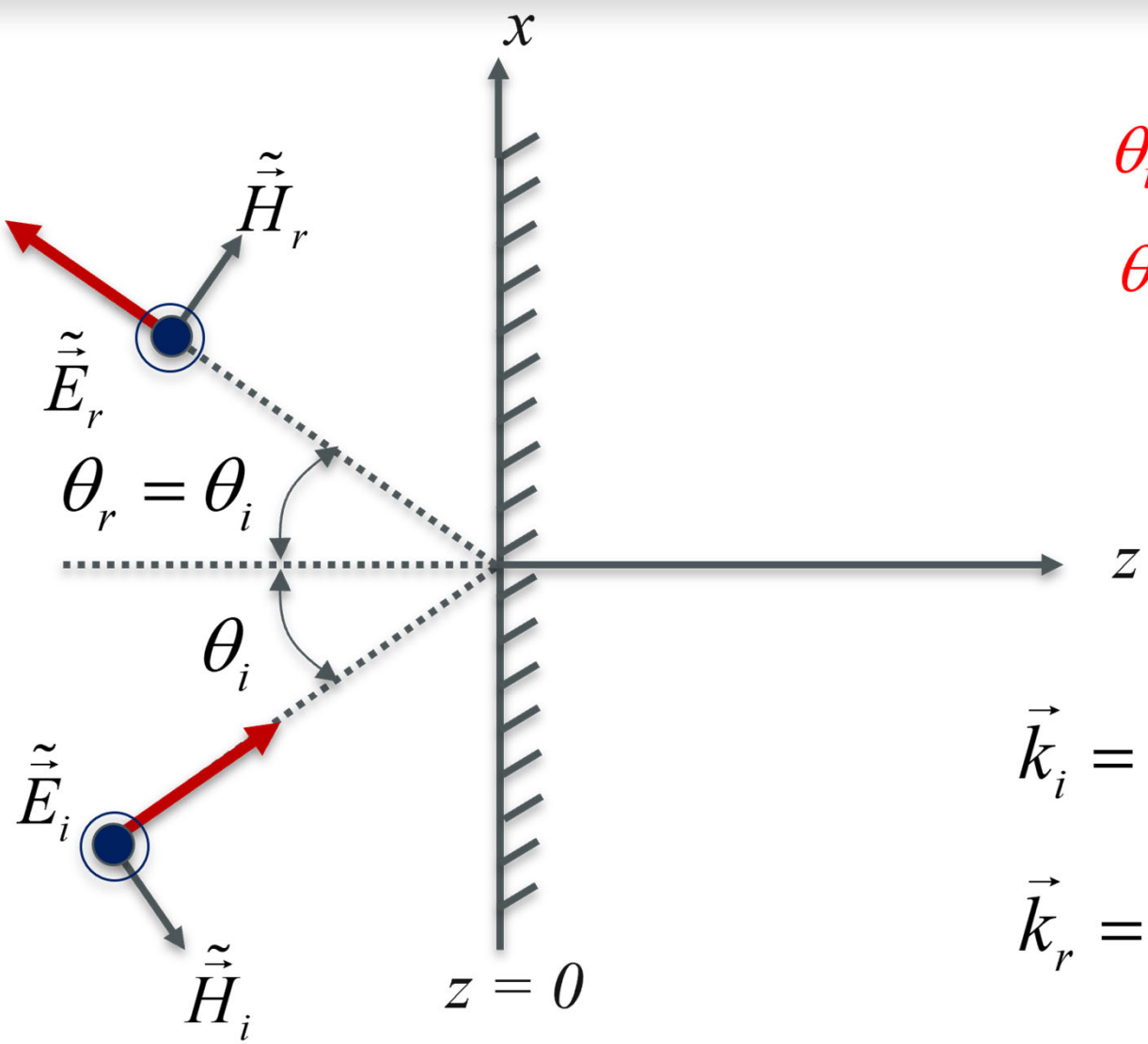
Effect of Polarization

- When a uniform plane wave is incident obliquely on a planar boundary between two distinct media, the behavior of the reflected wave depends on the polarization of the incident wave.
- The polarization of the incident wave is described with respect to the *plane of incidence*.
- The ***plane of incidence*** is the plane containing the direction of propagation and the normal to the boundary surface.
- Without loss of generality, we can always choose the plane of incidence to be the xz-plane (as in the sketch above).

Perpendicular and Parallel Polarization

- All possible polarization states of the incident plane wave can be described as a superposition of two cases:
  - *Perpendicular polarization*
  - *Parallel polarization*
- For ***perpendicular polarization***, the direction of the incident electric field is perpendicular to the plane of incidence.
- For ***parallel polarization***, the direction of the incident electric field is parallel to the plane of incidence.

Incident and Reflected Waves for Perpendicular Polarization



θ_i = angle of incidence

θ_r = angle of reflection

$$\vec{k}_i = k_1\left(\hat{x}\sin\theta_i + \hat{z}\cos\theta_i\right)$$

$$\vec{k}_r = k_1\left(\hat{x}\sin\theta_i - \hat{z}\cos\theta_i\right)$$

Incident Wave for Perpendicular Polarization

- The electric field of the incident wave is given by

$$\tilde{\vec{E}}_i = \hat{y}\, E_0^i\, e^{-j\vec{k}_i \bullet \vec{R}} = \hat{y}\, E_0^i\, e^{-jk_1\left(x\sin\theta_i + z\cos\theta_i\right)}$$

- The magnetic field of the incident wave is given by

$$\tilde{\vec{H}}_i = \frac{1}{\eta_1}\hat{k}_i \times \tilde{\vec{E}}_i = \frac{1}{\eta_1}\left(\hat{x}\sin\theta_i + \hat{z}\cos\theta_i\right)\times \tilde{\vec{E}}_i$$

$$= \left(-\hat{x}\cos\theta_i + \hat{z}\sin\theta_i\right)\frac{E_0^i}{\eta_1} e^{-jk_1\left(x\sin\theta_i + z\cos\theta_i\right)}$$

Reflected Wave for Perpendicular Polarization (1 of 2)

- Postulate that the electric field of the reflected wave is given by

$$\tilde{\vec{E}}_r = \hat{y}\, E_0^r\, e^{-j\vec{k}_r \bullet \vec{R}} = \hat{y}\, E_0^r\, e^{-jk_1\left(x\sin\theta_i - z\cos\theta_i\right)}$$

- The magnetic field of the reflected wave is given by

$$\tilde{\vec{H}}_r = \frac{1}{\eta_1}\hat{k}_r \times \tilde{\vec{E}}_r = \frac{1}{\eta_1}\left(\hat{x}\sin\theta_i - \hat{z}\cos\theta_i\right)\times \tilde{\vec{E}}_r$$

$$= \left(\hat{x}\cos\theta_i + \hat{z}\sin\theta_i\right)\frac{E_0^r}{\eta_1} e^{-jk_1\left(x\sin\theta_i - z\cos\theta_i\right)}$$

Reflected Wave for Perpendicular Polarization (2 of 2)

- To uniquely specify the reflected wave in terms of the incident wave, we need to evaluate $E_0{}^r$
- These parameters must be chosen such that the BC at $z = 0$ is satisfied.
- To satisfy the BC, the total electric field must be zero at $z = 0$ for all values of x.
- The above condition requires that:

$$E_0^r = -E_0^i$$

Reflected Wave for Perpendicular Polarization

- The electric field of the reflected wave is given by

$$\tilde{\vec{E}}_r = -\hat{y}E_0^i e^{-jk_1(x\sin\theta_i - z\cos\theta_i)}$$

- The magnetic field of the reflected wave is given by

$$\tilde{\vec{H}}_r = -(\hat{x}\cos\theta_i + \hat{z}\sin\theta_i)\frac{E_0^i}{\eta_1}e^{-jk_1(x\sin\theta_i - z\cos\theta_i)}$$

Total Field for Perpendicular Polarization (1 of 2)

- The total electric field is given by

$$\tilde{\vec{E}}_1 = \tilde{\vec{E}}_i + \tilde{\vec{E}}_r = -\hat{y} j 2E_0^i e^{-jk_1 x \sin\theta_i} \sin(k_1 z \cos\theta_i)$$

- The total magnetic field is given by

$$\begin{aligned}\tilde{\vec{H}}_1 &= \tilde{\vec{H}}_i + \tilde{\vec{H}}_r \\ &= -\frac{2E_0^i}{\eta_1} e^{-jk_1 x \sin\theta_i}[\hat{x}\cos\theta_i \cos(k_1 z \cos\theta_i) + \hat{z} j \sin\theta_i \sin(k_1 z \cos\theta_i)]\end{aligned}$$

Total Field for Perpendicular Polarization (2 of 2)

- Note that the total electric and magnetic fields can be described as standing waves in the z-direction because of the factors:

$$\cos(k_1 z \cos\theta_i)$$
$$\sin(k_1 z \cos\theta_i)$$

- The total electric and magnetic fields can be described as traveling waves in the x-direction because of the factor:

$$e^{-jk_1 x \sin\theta_i}$$

Standing Wave Pattern for Total Electric Field

- The nulls in the standing wave pattern for the total electric field occur when

$$\sin(k_1 z_m \cos\theta_i) = 0 \quad \Rightarrow k_1 z_m \cos\theta_i = m\pi, \quad m \in Z$$

$$z_m = \frac{-m\pi}{k_1 \cos\theta_i}, \quad m = 0,1,2,\cdots$$

$$z_m = \frac{-m\lambda}{2\cos\theta_i}, \quad m = 0,1,2,\cdots$$

Traveling Wave Behavior for Total Field (1 of 2)

- For the traveling wave factor, the 'effective' propagation constant can be obtained by considering

$$e^{-jk_1 x \sin\theta_i} = e^{-jk_{1x}x} \quad \Rightarrow k_{1x} = k_1 \sin\theta_i$$

- The phase velocity in the x-direction can be defined as

$$v_{1x} = \frac{\omega}{k_{1x}} = \frac{\omega}{k_1 \sin\theta_i}$$

Traveling Wave Behavior for Total Field (2 of 2)

- Notice that

$$v_{1x} = \frac{\omega}{k_1 \sin\theta_i} \geq v_1 = \frac{\omega}{k_1}$$

- If medium 1 were free space:

$$v_{1x} = \frac{\omega}{k_0 \sin\theta_i} \geq \frac{\omega}{k_0} = c$$

- **The phase velocity of the traveling wave exceeds the speed of light in the medium.**
- **No laws of physics are being violated since it is only 'information' that cannot travel faster than the speed of light.**
- **"Fast wave"**
- **This wave can be referred to as "transverse electric" ("TE") because the electric field is transverse to the direction of propagation (x) but the magnetic field is not.**

Summary

- The behavior of a plane wave obliquely incident on the planar boundary between a dielectric half-space and a perfect conductor depends on the polarization of the incident wave.
- The total fields exhibit a standing wave nature in the z-direction and traveling wave behavior in the x-direction.

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